Type MS · Read and write text files line by line

Answer 1 of 6 · Question 5 part c · 8 marks

2024 JUN · Paper 02 · extract 1 of 11 · find part c in this mark scheme extract

5(a)	A1 A2	Award **two** marks for a linked explanation such as: • Replacing individual variables with an array (using a single name) (1) means that comparisons could be done in a loop (1) • The animals could be processed in a loop (1) because they could be stored in an array (under a single name) (1)		**2**

Question	mp	Answer	Additional guidance	Mark

Type MS · Read and write text files line by line

Answer 1 of 6 · Question 5 part c · 8 marks

2024 JUN · Paper 02 · extract 2 of 11 · find part c in this mark scheme extract

5(b)		Award **one** mark for each of:	Ignore contents of fourth row if given.	
	B1	First row high index and low index (1) • 9, 0	Accept • 20, 10	
	B2	First row mid point (1) • 4	Accept • 15	
	B3	Second row (1) • 3, 0, 1	Accept • 14, 10, 12	
	B4	Third row (1) • 0, 0, 0	Accept • 10, 10, 10	**4**

Type MS · Read and write text files line by line

Answer 1 of 6 · Question 5 part c · 8 marks

2024 JUN · Paper 02 · extract 3 of 11 · find part c in this mark scheme extract

Example: Using index

highIndex	lowIndex	midPoint
9	0	4
3	0	1
0	0	0

Example: Using Values

highIndex	lowIndex	midPoint
20	10	15
14	10	12
10	10	10

Type MS · Read and write text files line by line

Answer 1 of 6 · Question 5 part c · 8 marks

2024 JUN · Paper 02 · extract 4 of 11 · find part c in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Read and write text files line by line

Answer 1 of 6 · Question 5 part c · 8 marks

2024 JUN · Paper 02 · extract 5 of 11 · find part c in this mark scheme extract

5(c)		Award **one** mark for each of:		
	C1	Opens the file for reading (1)		
	C2	Reads each line from the file (1)	Requires both read lines (prime, inside loop) for C# and Java. Requires a foreach for Python. Allow FT from C1	
	C3	Splits each line into individual strings (1)	Syntactically correct <anyString>.split(',') Does not need to be splitting correct line	
	C4	Individual values cast to int for processing (1)	Any method (index or foreach) allow float no FT allowed from C1 to C3 – must be full values between commas in file	
	C5	SubTotal += value **and** grandTotal += subTotal (1)	Any reasonable attempt at calculating both totals	
	C6	All given variable and constant names used throughout (1)	INPUT_FILE COMMA subTotal grandTotal	
	C7	Subtotals displayed / Grand total displayed (1)	Must be accurate as shown in QP Fig 8 Not hard-coded in any way	
	C8	The input file is closed, before exiting (1)	Syntactically correct Must be file name originally opened – FT from C1	**8**

Type MS · Read and write text files line by line

Answer 1 of 6 · Question 5 part c · 8 marks

2024 JUN · Paper 02 · extract 6 of 11 · find part c in this mark scheme extract

Q	mp	Code Examples

Type MS · Read and write text files line by line

Answer 1 of 6 · Question 5 part c · 8 marks

2024 JUN · Paper 02 · extract 7 of 11 · find part c in this mark scheme extract

C#

```
// Q05cFINISHED

using System.IO;

String inputFile = "Sales.txt";     // Output file name
String comma = ",";                 // Use as a constant

int subTotal = 0;               // Subtotal for each line
int grandTotal = 0;             // Running total
String line;                    // Line read from file

// Complete the code to open the file for reading
String fullPath = "C:\\Q05c";
inputFile = fullPath + "\\" + inputFile;
StreamReader theFile = new StreamReader (inputFile);

// Complete the code to read the first line of the input file
line = theFile.ReadLine ();

// Loop as long as the line is not null
while (line != null)
{
    // Complete the code to split the line into a set of five strings
    string[] stringSales = line.Split (comma);

    subTotal = 0;
    // Add code to sum up each value in the set of five strings
    foreach (string sale in stringSales)
    {
        subTotal = subTotal + Convert.ToInt32 (sale);
    }

    // Add code to display the subtotal for the line
    Console.WriteLine (Convert.ToString (subTotal));

    // Add code to calculate the running total
    grandTotal = grandTotal + subTotal;

    // Complete the code to read the next line of the file
    line = theFile.ReadLine ();
}

// Add code to display the total of all lines in the file
Console.WriteLine ("Grand total: " + Convert.ToString (grandTotal));

// Complete the code to close the opened file
theFile.Close ();

```

Type MS · Read and write text files line by line

Answer 1 of 6 · Question 5 part c · 8 marks

2024 JUN · Paper 02 · extract 8 of 11 · find part c in this mark scheme extract

Java

```java
// Q05cFINISHED

import java.io.FileReader;
import java.io.BufferedReader;

public class Q05cFINISHED
{
    public static void main(String[] args) throws Exception
    {
        String inputFile = "Sales.txt";     // Output file name
        String comma = ",";                  // Use as a constant

        int subTotal = 0;                // Subtotal for each line
        int grandTotal = 0;              // Running total
        String line;                     // Line read from file

        // Complete the code to open the file for reading
        String full_path = "C:\\Q05c";
        inputFile = full_path + "\\" + inputFile;
        BufferedReader theFile = new BufferedReader (new FileReader (inputFile));

        // Complete the code to read the first line of the input file
        line = theFile.readLine();

        // Loop as long as the line is not null
        while (line != null)
        {
            // Complete the code to split the line into a set of five strings
            String[] stringSales = line.split (comma);

            subTotal = 0;
            // Add code to sum up each value in the set of five strings
            for (int i = 0; i < stringSales.length; i++)
            {
                subTotal = subTotal + Integer.parseInt (stringSales[i]);
            }

            // Add code to display the subtotal for the line
            System.out.println (String.valueOf (subTotal));

            // Add code to calculate the running total
            grandTotal = grandTotal + subTotal;

            // Complete the code to read the next line of the file
            line = theFile.readLine();
        }

        // Add code to display the total of all lines in the file
        System.out.println ("Grand total: " + String.valueOf (grandTotal));

        // Complete the code to close the opened file
        theFile.close ();
    }
}
```

Type MS · Read and write text files line by line

Answer 1 of 6 · Question 5 part c · 8 marks

2024 JUN · Paper 02 · extract 9 of 11 · find part c in this mark scheme extract

Python

```python
# Q05cFINISHED

INPUT_FILE = "Sales.txt"    # Output file name
COMMA = ","                 # Use as a constant

subTotal = 0        # Subtotal for each line
grandTotal = 0      # Running total

# Complete the code to open the file for reading
theFile = open (INPUT_FILE, "r")

# Complete the code to read each line of the input file
for line in theFile:

    # Complete the code to split the line into a set of five strings
    stringSales = line.split (COMMA)

    subTotal = 0
    # Add code to sum up each value in the set of five strings
    for sale in stringSales:
        subTotal = subTotal + int (sale)

    # Add code to display the subtotal for the line
    print (str (subTotal))

    # Add code to calculate the running total
    grandTotal = grandTotal + subTotal

# Add code to display the total of all lines in the file
print ( "Grand total: " + str (grandTotal))

# Complete the code to close the opened file
theFile.close ()

```

Type MS · Read and write text files line by line

Answer 1 of 6 · Question 5 part c · 8 marks

2024 JUN · Paper 02 · extract 10 of 11 · find part c in this mark scheme extract

Type MS · Read and write text files line by line

Answer 1 of 6 · Question 5 part c · 8 marks

2024 JUN · Paper 02 · extract 11 of 11 · find part c in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Read and write text files line by line

Answer 2 of 6 · Question 3 part a · 6 marks

2019 JUN · Paper 02 · extract 1 of 17 · find part a in this mark scheme extract

Q3(b)	More memory/storage required (1) Because not sorted in place / not efficient / many steps / division of data set / is recursive (1)		(2)

Question	Answer	Additional Guidance	Mark

Type MS · Read and write text files line by line

Answer 2 of 6 · Question 3 part a · 6 marks

2019 JUN · Paper 02 · extract 2 of 17 · find part a in this mark scheme extract

Q3(a)	Award 1 mark each for: 1. File opened for input (1) 2. File opened for output (1) 3. Loop for processing data (1) 4. Check for '@' symbol (1) 5. Write invalid address to output file (1) 6. Close files (1)	Amended code should be saved as Q03aFINISHED. Do not penalise candidates who attempt more than the stated requirements.	(6)

Type MS · Read and write text files line by line

Answer 2 of 6 · Question 3 part a · 6 marks

2019 JUN · Paper 02 · extract 3 of 17 · find part a in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Read and write text files line by line

Answer 2 of 6 · Question 3 part a · 6 marks

2019 JUN · Paper 02 · extract 4 of 17 · find part a in this mark scheme extract

Q3(c)	Award 1 mark for each section of code that: 1. Asks for input of a number and stores it in a variable with a meaningful name (1) 2. Checks number is between 1 and 50 (1) 3. Calculates square and cube of number (1) 4. Displays suitable labels e.g. 'Number', 'Square' and 'Cube' and relevant values (1) For the whole code: 5. Compiles and runs without logical or syntax errors (1) 6. Efficient in terms of computation, storage and selection of programming constructs (1)	Amended code should be saved as Q03cFINISHED. Allow follow through for mark point 4 when square and cube are incorrectly calculated. Credit alternative coding which produces same results. Accept use of exponent operators: C# e.g. Math.Pow (2, 3) = 8 Java e.g. Math.pow (2, 3) = 8 Python e.g. 2**3 = 8.	(6)

Question	Designated marks:	Additional Guidance	Mark

Type MS · Read and write text files line by line

Answer 2 of 6 · Question 3 part a · 6 marks

2019 JUN · Paper 02 · extract 5 of 17 · find part a in this mark scheme extract

Java Example code

Type MS · Read and write text files line by line

Answer 2 of 6 · Question 3 part a · 6 marks

2019 JUN · Paper 02 · extract 6 of 17 · find part a in this mark scheme extract

3(a)

```
package q03aFINISHED;

import java.io.BufferedReader;
import java.io.FileNotFoundException;
import java.io.FileReader;
import java.io.PrintWriter;
import java.io.UnsupportedEncodingException;
import java.util.Scanner;

public class Q03aFINISHED
{
    public static void main(String[] args) throws FileNotFoundException,UnsupportedEncodingException
    {
        String email="";
        PrintWriter errorFile;

        //  Open the input file

        Scanner emailFile = new Scanner(new BufferedReader(new FileReader("Email.txt")));

        //  Open the output file

        errorFile = new PrintWriter("Error.txt", "UTF-8");

        //  Find errors and write to output file

        while (emailFile.hasNextLine())
        {
            email = emailFile.nextLine();
            if (!email.contains("@"))
            {
                errorFile.println(email);
            }
        }

        //  Close files

        emailFile.close();
        errorFile.close();
    }
}
```

Type MS · Read and write text files line by line

Answer 2 of 6 · Question 3 part a · 6 marks

2019 JUN · Paper 02 · extract 7 of 17 · find part a in this mark scheme extract

3(c)	Note: Line 19 is an alternative response to line 18
	
	`package q03cFINISHED;`
	
	`import java.util.Scanner;`
	
	`public class Q03c`
	`{`
	`  public static void main(String[] args)`
	`  {`
	`  // Write your code here`
	`  Scanner input = new Scanner(System.in);`
	`  int myValue = 1;`
	`  while (myValue >= 1 && myValue <= 50)`
	`  {`
	`  System.out.println("Enter a value from 1 to 50: ");`
	`  myValue = input.nextInt();`
	`  if (myValue >= 1 && myValue <= 50)`
	`  {`
	`  System.out.println("Value " + myValue + " Square " + myValue * myValue + " Cube " + myValue * myValue * myValue);`
	`  System.out.println("Value " + myValue + " Square " + Math.pow(myValue, 2) + " Cube " + Math.pow(myValue, 3));`
	`  }`
	`  }`
	`  }`
	`}`	

Type MS · Read and write text files line by line

Answer 2 of 6 · Question 3 part a · 6 marks

2019 JUN · Paper 02 · extract 8 of 17 · find part a in this mark scheme extract

5

```
        int total = 0;
        double average;
        int books[]  = new int[libraryRecord.length];
        int gold = 0, silver = 0, bronze = 0;
        int goldID = 0, silverID = 0, bronzeID = 0;

        System.out.println("Pupil ID of pupils who have read less than 10 books");

            for (int i = 0; i < libraryRecord.length; i++)
            {
                books[i] = Integer.parseInt(libraryRecord[i][3]);
                if (books[i] > gold)
                {
                    bronze = silver; silver = gold; gold = books[i]; goldID = i;
                }
                else if (books[i] > silver)
                {
                    bronze = silver; silver = books[i]; silverID = i;
                }
                else if (books[i] > bronze)
                {
                    bronze = books[i]; bronzeID = i;
                }
                total = total + books[i];

                if (books[i] < 10)
                {
                    System.out.println(libraryRecord[i][0]);
                }
            }
            System.out.println();

            System.out.println("Total books read by pupils = " + total);

            System.out.println();

            average = total / libraryRecord.length;

            System.out.println("Average books read by pupils = " + average);

            System.out.println();

            System.out.println("Gold winner is " + libraryRecord[goldID][1] + " " + libraryRecord[goldID][2]);
            System.out.println("Silver winner is " + libraryRecord[silverID][1] + " " + libraryRecord[silverID][2]);
            System.out.println("Bronze winner is " + libraryRecord[bronzeID][1] + " " + libraryRecord[bronzeID][2]);
```

Type MS · Read and write text files line by line

Answer 2 of 6 · Question 3 part a · 6 marks

2019 JUN · Paper 02 · extract 9 of 17 · find part a in this mark scheme extract

Python Example code

1(c)

```
# Q01c

count = 0

maxValue = 5
constantValue = 7

while (count < maxValue):
    print(count, count + constantValue)
    count = count + 1
```

1(f)

```
    if (shopIncome > 5000) or (assistantSales >= 10):
        print ("Assistant bonus = ", assistantSalary * 0.1)

    elif (shopIncome >= 2000) and (assistantSales >= 5):
        print("Assistant bonus = ", assistantSalary * 0.05)
```

Type MS · Read and write text files line by line

Answer 2 of 6 · Question 3 part a · 6 marks

2019 JUN · Paper 02 · extract 10 of 17 · find part a in this mark scheme extract

2(a)

```
# Initialise variables
counter = 1
answer = randint(1,10)
guess = 0

# Print prompt and take guess from user
guess = int(input("Enter a number from 1 to 10: "))

# Create while loop to check answer
while guess!=answer:
    counter = counter + 1
    if(guess > answer):
        print (guess, " was too high. Try again.")
    else:
        print (guess, " was too low. Try again.")

    guess = int(input("Guess again: "))

# Report the correct answer to the user and indicate the number of guesses
print ("You guessed", guess, "in", counter, "guesses")
```

Type MS · Read and write text files line by line

Answer 2 of 6 · Question 3 part a · 6 marks

2019 JUN · Paper 02 · extract 11 of 17 · find part a in this mark scheme extract

3(a)

```
Q03aFINISHED

#   Open file and input data
with open('Email.txt','r') as inputFile:
    emailList = inputFile.readlines()

#   Open file for output
outputFile = open('Error.txt', 'a')

#   Find errors and write to error log
for address in emailList:
    if not "@" in address:
        outputFile.write(address)

#   Close files
outputFile.close()
inputFile.close()
```

Type MS · Read and write text files line by line

Answer 2 of 6 · Question 3 part a · 6 marks

2019 JUN · Paper 02 · extract 12 of 17 · find part a in this mark scheme extract

3(c)

```
1  # Q03c
2
3  # Write your code below this line
4  myValue = 1
5
6  while (myValue >= 1 and myValue <= 50):
7      myValue = int(input("Enter a value from 1 to 50: "))
8      if (myValue >= 1 and myValue <= 50):
9          print("Value: ", myValue, "Square: ",myValue*myValue, "Cube: ", myValue*myValue*
           myValue)
10
11         print("Value: ", myValue, "Square: ", myValue**2, "Cube: ", myValue**3)
```

Note: Line 11 is an alternative response to line 9.

Type MS · Read and write text files line by line

Answer 2 of 6 · Question 3 part a · 6 marks

2019 JUN · Paper 02 · extract 13 of 17 · find part a in this mark scheme extract

5

```
# Write your code below this line

total = 0
gold = 0
silver = 0
bronze = 0
goldID = 0
silverID = 0
bronzeID = 0

print ("Pupil ID of pupils who have read less than 10 books ")

for row in range(len(libraryRecord)):
    books = int(libraryRecord[row][3])
    if (books > gold):
        bronze = silver
        silver = gold
        gold = books
        goldID = row
    elif (books > silver):
        bronze = silver
        silver = books
        silverID = row
    elif (books > bronze):
        bronze = books
        bronzeID = row

    total = total + books

    if (books < 10):
        print ("\t",libraryRecord[row][0])

print ("\n")
print ("Total number of books read by pupils: " , total)

print ("\n")
print ("Average number of books read by pupils: " , total / len(libraryRecord[0]))

print ("\n")
print ("Gold medal winner is: \t\t", libraryRecord[goldID][1], " ", libraryRecord[goldID][2])
print ("Silver medal winner is: \t", libraryRecord[silverID][1], " ", libraryRecord[silverID][2])
print ("Bronze medal winner is: \t", libraryRecord[bronzeID][1], " ", libraryRecord[bronzeID][2])
```

Type MS · Read and write text files line by line

Answer 2 of 6 · Question 3 part a · 6 marks

2019 JUN · Paper 02 · extract 14 of 17 · find part a in this mark scheme extract

Type MS · Read and write text files line by line

Answer 2 of 6 · Question 3 part a · 6 marks

2019 JUN · Paper 02 · extract 15 of 17 · find part a in this mark scheme extract

Type MS · Read and write text files line by line

Answer 2 of 6 · Question 3 part a · 6 marks

2019 JUN · Paper 02 · extract 16 of 17 · find part a in this mark scheme extract

Type MS · Read and write text files line by line

Answer 2 of 6 · Question 3 part a · 6 marks

2019 JUN · Paper 02 · extract 17 of 17 · find part a in this mark scheme extract

Pearson Education Limited. Registered company number 872828
with its registered office at 80 Strand, London, WC2R 0RL, United Kingdom

Type MS · Read and write text files line by line

Answer 3 of 6 · Question 3 part a · 8 marks

2021 JUN · Paper 02 · extract 1 of 8 · find part a in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
3 (b)	B1	Award up to 2 marks for a linked explanation such as: • The number of keys are limited/only one shift used (1) making it easy to use brute force to decrypt (1) • It can be easy to find commonly used letters (e.g. E) (1) and guess the key (1)	Accept alternative similar wording.	(2)

Type MS · Read and write text files line by line

Answer 3 of 6 · Question 3 part a · 8 marks

2021 JUN · Paper 02 · extract 2 of 8 · find part a in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
3 (c)		Award 1 mark each up to a maximum of 4 for: <table><tr><td>Encrypted letter</td><td>f</td><td>l</td><td>m</td><td>k</td><td>t</td><td>r</td><td>w</td><td>h</td><td>e</td><td>e</td></tr><tr><td>Keyword letter</td><td>t</td><td>h</td><td>i</td><td>r</td><td>t</td><td>y</td><td>t</td><td>h</td><td>i</td><td>r</td></tr><tr><td>Decrypted letter</td><td>m</td><td>e</td><td>e</td><td>t</td><td>a</td><td>t</td><td>d</td><td>a</td><td>w</td><td>n</td></tr></table>		
	C1	Award 1 mark from: • flmktrwhee mapped to thirtythir (1) • Flmktr mapped to thirty (1)		
	C2	At least one letter decrypted correctly (1)		
	C3	At least one word decrypted correctly (1)		
	C4	Decrypted message 'meet at dawn' (1)		(4)

Type MS · Read and write text files line by line

Answer 3 of 6 · Question 3 part a · 8 marks

2021 JUN · Paper 02 · extract 3 of 8 · find part a in this mark scheme extract

Question		Answer	Additional Guidance	Mark
3 (d)(i)	D1	Award 1 mark for: • cipher/ letter / a single encrypted letter (1)	Do not accept word/message/text	(1)

Type MS · Read and write text files line by line

Answer 3 of 6 · Question 3 part a · 8 marks

2021 JUN · Paper 02 · extract 4 of 8 · find part a in this mark scheme extract

		• cipherLetter / a single encrypted letter (1)		(1)
3 (d)(ii)	D2	Award 1 mark for any of: • keywordLetter • plaintextLetter	Ignore case	(1)

Type MS · Read and write text files line by line

Answer 3 of 6 · Question 3 part a · 8 marks

2021 JUN · Paper 02 · extract 5 of 8 · find part a in this mark scheme extract

		• plaintextLetter		(1)
3 (d)(iii)	D2	Award 1 mark for any of: • subprogram that is already defined • subprogram that is already written • subprogram that is already compiled • subprogram that can be called without having to write code for it	Must be clear they are referring to subprogram provided by the language itself	(1)

Type MS · Read and write text files line by line

Answer 3 of 6 · Question 3 part a · 8 marks

2021 JUN · Paper 02 · extract 6 of 8 · find part a in this mark scheme extract

Type MS · Read and write text files line by line

Answer 3 of 6 · Question 3 part a · 8 marks

2021 JUN · Paper 02 · extract 7 of 8 · find part a in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
3 (a)	A1	Get plaintext and store in plaintext variable	Accept alternative wording Line numbers may be different compared to the examples shown When testing the completed code use lowercase for the input	(1)
	A2	Get key and store in key variable		(1)
	A3	Validate key (must repeat until a valid key has been input)		(1)
	A4	Open file to write/append		(1)
	A5	Write cipher text		(1)
	A6	Close file		(1)
	A7	Displays ciphertext		(1)
	A8	Executing and producing correct output to file and screen (must have A3)		(1)
Code examples				
Python				

```
15
16 #       Add your code to get the plaintext and convert it to lowercase
17 plaintext = input("Enter the plaintext using lowercase letters: ")
18 plaintext = plaintext.lower()
```

```
# Add your code to get the key and make sure the key is between 1 and 25
while key < 1 or key > 25:
    key = int(input("Enter the key - a number between 1 and 25 "))
```

```
39 # Add your code to write the ciphertext to a text file
40 cipherFile = open("Cipher.txt", "w")
41 cipherFile.write(ciphertext)
42 cipherFile.close()
```

```
40
41 #       Add your code to display the ciphertext
42 print("The ciphertext message is:",ciphertext)
43
```

Type MS · Read and write text files line by line

Answer 3 of 6 · Question 3 part a · 8 marks

2021 JUN · Paper 02 · extract 8 of 8 · find part a in this mark scheme extract

Type MS · Read and write text files line by line

Answer 4 of 6 · Question 5 part a · 9 marks

2022 JUN · Paper 02 · extract 1 of 9 · find part a in this mark scheme extract

5 (b)(i)	B1 B2 B3	**Award one mark for each:** • Pass 1 correct in first row of the response (1) • Pass 2 **OR** 3 correct in any row of the response (1) • All passes correct and no extra passes (1) 2 3 8 1 9 10 2 3 1 8 9 10 2 1 3 8 9 10 1 2 3 8 9 10	If the final row (1, 2, 3, 8, 9, 10) is repeated treat as the same row.	(3)

Type MS · Read and write text files line by line

Answer 4 of 6 · Question 5 part a · 9 marks

2022 JUN · Paper 02 · extract 2 of 9 · find part a in this mark scheme extract

5 (b)(ii)	B4 B5	**Award up to 2 marks for a linked explanation such as:** • The sorting is done in the same place as the original data (1), which means only one additional variable is needed to store the value being swapped (when the values are out of order (1) • All sort and swapping operations are done on the original data (1) which means the amount of memory needed is constant (1) • The list does not need to be split (1), which would use more memory (1)		(2)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Read and write text files line by line

Answer 4 of 6 · Question 5 part a · 9 marks

2022 JUN · Paper 02 · extract 3 of 9 · find part a in this mark scheme extract

5 (c)	C1 C2 C3	**Award up to 3 marks for a linked description that includes:** • Start at the beginning of the list (1) • Compare each value in the list with the search item (1) • Repeat until a match is found / the end of the list is reached (1)		(3)

Type MS · Read and write text files line by line

Answer 4 of 6 · Question 5 part a · 9 marks

2022 JUN · Paper 02 · extract 4 of 9 · find part a in this mark scheme extract

Code - Python

Question	mp	Answer	Additional Guidance	Mark

Type MS · Read and write text files line by line

Answer 4 of 6 · Question 5 part a · 9 marks

2022 JUN · Paper 02 · extract 5 of 9 · find part a in this mark scheme extract

5 (a)	**Award 1 mark for each of:**			
	A1	Variable to store a number of incorrect passwords (1)		
	A2	Loop used (1)		
	A3	Each password checked to see if the first character is an uppercase letter (1)		
	A4	Each password checked to see if it contains a digit (1)		
	A5	Only check for digits if first character is uppercase OR Only checks if first character is uppercase if at least one digit is present (1)		
	A6	Number of incorrect passwords incremented correctly (1) (allow follow through)		
	A7	All incorrect passwords displayed (1) (allow follow through but must try to display those without uppercase and those without digits)		
	A8	Text file closed (1)		
	A9	Number of incorrect passwords displayed (1) (allow follow through)		(9)
Code examples				
Python				

```
# Add your code here
incorrectPasswords = 0

for password in passwordFile:
    valid = False

    if password[0] in alphabet:
        index = 1
        while (valid == False) and (index in range (1, len(password))):
            if password[index]in digit:
                valid = True
            else:
                index = index + 1

    if not valid:
        incorrectPasswords = incorrectPasswords + 1
        print(password)

print(incorrectPasswords,"passwords are incorrect.")

passwordFile.close()
```

Type MS · Read and write text files line by line

Answer 4 of 6 · Question 5 part a · 9 marks

2022 JUN · Paper 02 · extract 6 of 9 · find part a in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Read and write text files line by line

Answer 4 of 6 · Question 5 part a · 9 marks

2022 JUN · Paper 02 · extract 7 of 9 · find part a in this mark scheme extract

Code examples

Python

Subprograms

```
# Add your subprograms here
def checkInput(pWordLength):
    # Validate input to make sure it is the same length as the secret word
    check = ""
    while len(check) != pWordLength:
        check = input("Enter your guess. The secret word has " +str(pWordLength)+ " letters: ")
    return(check)

def checkLettersInWord(pGuess, pWordToGuess, pLetters, type):
    for letter in pGuess: # Check each letter in the guess
        if type == 0: # The check to carry out is for correct letters
            # If the letter is in the secret word and not already in the correctletters, add it
            if letter in pWordToGuess and letter not in pLetters:
                pLetters = pLetters + letter
        else: # The check to carry out is for wrong letters
            # If the letter is not in the secret word and not already in the wrongLetters, add
            if letter not in pWordToGuess and letter not in pLetters:
                pLetters = pLetters + letter
    return pLetters

def display(pCorrectLetters, pWrongLetters):
    # Display the correct and wrong letters
    if len(pCorrectLetters) > 0:
        print("Letter(s) in the secret word:",pCorrectLetters)

    if len(pWrongLetters) > 0:
        print("Letter(s) not in the secret word:",pWrongLetters)
```

Type MS · Read and write text files line by line

Answer 4 of 6 · Question 5 part a · 9 marks

2022 JUN · Paper 02 · extract 8 of 9 · find part a in this mark scheme extract

Main program

```
# Add your main program code here
wordLength = len(wordToGuess)
maxAttempts = wordLength + 3
numAttempts = 0
correctLetters = ""
wrongLetters = ""
guessed = False

# Loop until the no attempts are left or the secret word has been guessed
while numAttempts < maxAttempts and guessed == False:
    print("You have",maxAttempts - numAttempts,"attempts to guess the secret word.")
    numAttempts = numAttempts + 1

    guess = checkInput(wordLength)  # Validate the input

    if guess == wordToGuess: # Check to see if the secret word has been guessed
        guessed = True
        print("Well done. You guessed the secret word was",wordToGuess,"in",numAttempts,"attempts.")

    else: # If it hasn't been guessed check for correct and wrong letters
        correctLetters = checkLettersInWord(guess,wordToGuess, correctLetters, 0)
        wrongLetters = checkLettersInWord(guess,wordToGuess, wrongLetters, 1)
        display(correctLetters, wrongLetters)

# Display game over message if attempts have run out and the secret word has not been guessed
if numAttempts == maxAttempts and guessed == False:
    print("Game over. You did not guess that the secret word was",wordToGuess)
```

Type MS · Read and write text files line by line

Answer 4 of 6 · Question 5 part a · 9 marks

2022 JUN · Paper 02 · extract 9 of 9 · find part a in this mark scheme extract

Pearson Education Limited. Registered company number 872828
with its registered office at 80 Strand, London, WC2R 0RL, United Kingdom

Type MS · Read and write text files line by line

Answer 5 of 6 · Question 5 part c · 6 marks

2023 JUN · Paper 02 · extract 1 of 9 · find part c in this mark scheme extract

5(a)	A1 A2	Award 1 mark for demonstration of one correctly merged/split row Award 2 marks for correct merge sort Blackfin, Bigeye | Longtail, Albacore | Bluefin Longtail, Blackfin, Bigeye, Albacore | Bluefin Longtail, Bluefin, Blackfin, Bigeye, Albacore OR Bigeye | Blackfin, Albacore | Longtail, Bluefin Bigeye | Longtail, Bluefin, Blackfin, Albacore Longtail, Bluefin, Blackfin, Bigeye, Albacore OR Blackfin, Bigeye | Albacore | Longtail, Bluefin Blackfin, Bigeye | Longtail, Bluefin, Albacore	We are testing pupils understanding of how a merge sort progresses: • Divide and conquer (split) • Merging adjacent elements Allow answers in ascending order	**2**

Type MS · Read and write text files line by line

Answer 5 of 6 · Question 5 part c · 6 marks

2023 JUN · Paper 02 · extract 2 of 9 · find part c in this mark scheme extract

		Longtail, Bluefin, Blackfin, Bigeye, Albacore OR Blackfin, Bigeye | Albacore | Longtail, Bluefin Blackfin, Bigeye, Albacore | Longtail, Bluefin Longtail, Bluefin, Blackfin, Bigeye, Albacore		

Type MS · Read and write text files line by line

Answer 5 of 6 · Question 5 part c · 6 marks

2023 JUN · Paper 02 · extract 3 of 9 · find part c in this mark scheme extract

<table>
<tr><th>Question</th><th>mp</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(b)</td><td>B1
B2</td><td>Award one mark for each correct cell:
<table><tr><td>Line number with error</td><td>11 (1)</td></tr><tr><td>Corrected line of pseudocode</td><td>SET tmp TO myTuna[ndx] (1)</td></tr></table></td><td>Ignore spellings

Ignore missing text as long as the minimum of [ndx] is provided

B2 does not depend on B1</td><td>2</td></tr>
</table>

Question	mp	Answer	Additional guidance	Mark
5(c)	Award **one** mark for each of:		C1 allow append as writing to file C4 Do not award comma within an array C5 must have 1 complete line written to file (allow additional spaces and commas). Should include individual elements (2D index) and not a 1D array from tbl_tuna C6 complete file must be as expected e.g. no additional comma at end of line and no spaces around commas	**6**

Type MS · Read and write text files line by line

Answer 5 of 6 · Question 5 part c · 6 marks

2023 JUN · Paper 02 · extract 4 of 9 · find part c in this mark scheme extract

C#

```
// Q05cFINISHED

using System.IO;

String[,] tblTuna =  {
    { "Yellowfin", "105", "15", "3"},
    { "Albacore","90","15","5"},
    { "Skipjack","50","3","4"},
    { "Bigeye","105","25","4"},
    { "Atlantic Bonito","50","4","2"},
    { "Northern Bluefin","190","120","11"},
    { "Southern Bluefin","190","120","11"},
    { "Tongol","90","20","4"}
};



// ------------------------------------------
// Write your code below this line

String fileName = "TunaData.txt";       // Output file name
String comma = ",";                     // Use as a constant
String full_path = "C:\\Q05c";
fileName = full_path + "\\" + fileName;

// Create variables as needed
int number = 101;           // Number for the line at the front
String lineOut = "";        // The line to output

// Get the dimensions of the array for looping
int rows = tblTuna.GetLength(0);
int columns = tblTuna.GetLength(1);

// Open the file for writing
StreamWriter fileWriter = new StreamWriter(fileName);

```

Type MS · Read and write text files line by line

Answer 5 of 6 · Question 5 part c · 6 marks

2023 JUN · Paper 02 · extract 5 of 9 · find part c in this mark scheme extract

```
// Process every tuna in the table
for (int i = 0; i < rows; i++)          // Rows
{
    // Start with the number and a comma
    lineOut = number.ToString() + comma;

    for (int j = 0; j < columns; j++)   // Columns
    {
        // Add the name and the numbers
        lineOut = lineOut + tblTuna[i, j];

        // Add a comma to all the fields except the last
        if (j < columns - 1)
        {
            lineOut = lineOut + comma;
        }
    }
    // Write the line to the file
    fileWriter.WriteLine(lineOut);

    // Go to the next number for the line
    number = number + 1;
}

// Close the file
fileWriter.Close();
```

Type MS · Read and write text files line by line

Answer 5 of 6 · Question 5 part c · 6 marks

2023 JUN · Paper 02 · extract 6 of 9 · find part c in this mark scheme extract

Java

```java
// Q05cFINISHED

import java.io.FileWriter;
import java.io.IOException;

public class Q05cFINISHED {

    public static void main(String[] args) throws Exception {

        String[][] tblTuna =
        {
            { "Yellowfin", "105", "15", "3"},
            { "Albacore","90","15","5"},
            { "Skipjack","50","3","4"},
            { "Bigeye","105","25","4"},
            { "Atlantic Bonito","50","4","2"},
            { "Northern Bluefin","190","120","11"},
            { "Southern Bluefin","190","120","11"},
            { "Tongol","90","20","4"}
        };

        // ------------------------------------------
        // Write your code below this
        String fileName = "TunaData.txt";       // Output file name
        String comma = ",";                     // Use as a constant
        String full_path = "C:\\src\\q05c";
        fileName = full_path + "\\" + fileName;

        // Create variables as needed
        int number = 101;           // Number for the line at the front
        String lineOut = "";        // The line to output
        String linefeed = "\n";     // Use as a constant

        // Get the dimensions of the array for looping
        int rows = tblTuna.length;
        int columns = tblTuna[0].length;

        // Open the file for writing
        FileWriter fileWriter = new FileWriter(fileName);

```

Type MS · Read and write text files line by line

Answer 5 of 6 · Question 5 part c · 6 marks

2023 JUN · Paper 02 · extract 7 of 9 · find part c in this mark scheme extract

```
41          // Process every tuna in the table
42          for (int i = 0; i < rows; i++)          // Rows
43          {
44              // Start with the number and a comma
45              lineOut = Integer.toString(number) + comma;
46
47              for (int j = 0; j < columns; j++)   // Columns
48              {
49                  // Add the name and the numbers
50                  lineOut = lineOut + tblTuna[i][j];
51
52                  // Add a comma to all the fields except the last
53                  if (j < columns - 1)
54                  {
55                      lineOut = lineOut + comma;
56                  }
57                  else
58                  {
59                      lineOut = lineOut + linefeed;
60                  }
61              }
62              // Write the line to the file
63              fileWriter.write(lineOut);
64
65              // Go to the next number for the line
66              number = number + 1;
67          }
68
69          // Close the file
70          fileWriter.close();
71      }
72
73  }
```

Type MS · Read and write text files line by line

Answer 5 of 6 · Question 5 part c · 6 marks

2023 JUN · Paper 02 · extract 8 of 9 · find part c in this mark scheme extract

Python

```
# Q05cFINISHED

tbl_tuna = [["Yellowfin",105,15,3],
            ["Albacore",90,15,5],
            ["Skipjack",50,3,4],
            ["Bigeye",105,25,4],
            ["Atlantic Bonito",50,4,2],
            ["Northern Bluefin",190,120,11],
            ["Southern Bluefin",190,120,11],
            ["Tongol",90,20,4]]

# ---------------------------------------------
# Write your code below this line

FILE_OUT = "TunaData.txt"           # Output file name
COMMA = ","                         # Use as constant

# Create variables as needed
LINEFEED = "\n"
number = 101                        # Number at the front
line_out = ""                       # The line to write

# Open the file for writing
file = open (FILE_OUT, "w")

# Process every tuna in the table
for tuna in tbl_tuna:
    # Start with the number and a comma
    line_out = str (number) + COMMA

    # Add the name of the tuna and a comma
    line_out = line_out + tuna[0] + COMMA

    # Each of the numbers need to be converted to a string
    # before adding to the output string.
    line_out = line_out + str (tuna[1]) + COMMA
    line_out = line_out + str (tuna[2]) + COMMA
    line_out = line_out + str (tuna[3])     # No comma on last field

    # Add a line feed to the whole line
    line_out = line_out + LINEFEED

    # Write the line to the file
    file.write (line_out)

    # Go to the next number for the line
    number = number + 1

# Close the file
file.close ()
```

Type MS · Read and write text files line by line

Answer 5 of 6 · Question 5 part c · 6 marks

2023 JUN · Paper 02 · extract 9 of 9 · find part c in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Read and write text files line by line

Answer 6 of 6 · Question 5 part b · 8 marks

2025 JUN · Paper 02 · extract 1 of 4 · find part b in this mark scheme extract

5(a)(i)	A1 A2	final score not accurate (1) because score is resetting/score keeps going back to zero (in each iteration of the loop) (1)	'Score not accurate' is not enough on its own 'Score on line 20' or 'Score output to DISPLAY' would count as final score.	**2**

Question	mp	Answer	Additional guidance	Mark
5(a)(ii)	A3 A4	Award up to **two** marks for a linked description such as: • Asking 10 addition questions (1) and calculating final score (1) • Asking randomised adding questions (1) and checking user answer against computer answer (1) • Creating adding quiz with random numbers up to 99 (1) and outputting number of correct answers (1)	Processes that can be included Concept of this being an 'Adding Quiz' required • Getting random numbers for question • Calculating sum of two random numbers for correct answers • comparing user answer to correct answer • incrementing /tracking score • asking 10 questions (loop) • outputting final score	**2**

17 of 23

Type MS · Read and write text files line by line

Answer 6 of 6 · Question 5 part b · 8 marks

2025 JUN · Paper 02 · extract 2 of 4 · find part b in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
5(b)				
	B1	Open file for writing (1)	"w" mode only Accept "w+"	
	B2	Selection used to get grade (1)	Must address all 4 grades Does not have to be accurate on boundaries e.g. >80 is ok for distinction etc.1	
	B3	Output suitable message to say file created (1)		
	B4	Evidence of 2D indexing (1)	Even if element not used in rest of program 2D indexing **or** 1D index within *for-each* structure **or** sum(1Darray) or min(1Darray) etc. At least 1 inner value accessed	
	B5	Lowest score **must** be found and omitted/subtracted from calculated scores (all arrays) (1)	Any method e.g.: • List methods • Selection statements • iterable functions must be done programmatically	
	B6	At least 1 line written to file (1)	Must contain something generated by program (total/result), does not have to be correct total or grade	
	B7	Add line feed to write line after both score and grade (1)		
	B8	Output file all correct (1)		**8**

18 of 23

Type MS · Read and write text files line by line

Answer 6 of 6 · Question 5 part b · 8 marks

2025 JUN · Paper 02 · extract 3 of 4 · find part b in this mark scheme extract

Paper 2 – Application of Computational Thinking

```
# Q05b

Tbl_scores = [[45,42,43],[19,36,41],[48,48,48],[35,32,31],
              [18,22,26],[41,35,31],[15,19,12],[41,39,40],
              [27,25,22],[32,36,35],[37,36,37],[40,35,41]]

# ----------------------------------------------
# Write your code below this line
total = 0
grade = ""
outString = ""

with open ("grades.txt","w") as theFile:

    # calculate each students score
    for student in Tbl_scores:
        student.sort()
        total = student[1] + student[2]

        # decide student grade
        if total >= 80:
            grade = "Distinction"
        elif total >= 65:
            grade = "Merit"
        elif total >= 50:
            grade = "Pass"
        else:
            grade = "Fail"

        # write score and grade to file
        outString = str(total) + " " + grade + "\n"
        theFile.write(outString)

print ("Grade file has been created")
```

```
# Q05b

Tbl_scores = [[45,42,43],[19,36,41],[48,48,48],[35,32,31],
              [18,22,26],[41,35,31],[15,19,12],[41,39,40],
              [27,25,22],[32,36,35],[37,36,37],[40,35,41]]

# ----------------------------------------------
# Write your code below this line
total = 0
grade = ""
outString = ""

theFile = open ("grades.txt","w")

# calculate each students score
for student in Tbl_scores:
    total = sum(student) - min(student)

    # decide student grade
    if total >= 80:
        grade = "Distinction"
    elif total >= 65:
        grade = "Merit"
    elif total >= 50:
        grade = "Pass"
    else:
        grade = "Fail"

    # write score and grade to file
    outString = str(total) + " " + grade + "\n"
    theFile.write(outString)

theFile.close()

print ("Grade file has been created")
```

```
# Q05b

Tbl_scores = [[45,42,43],[19,36,41],[48,48,48],[35,32,31],
              [18,22,26],[41,35,31],[15,19,12],[41,39,40],
              [27,25,22],[32,36,35],[37,36,37],[40,35,41]]

# ----------------------------------------------
# Write your code below this line
grade = ""
outString = ""

theFile = open ("grades.txt","w")

# calculate each students score
for student in Tbl_scores:
    total = 0
    lowest = 51
    for score in student:
        if score < lowest:
            lowest = score
        total = total + score
    total = total - lowest

    # decide student grade
    if total >= 80:
        grade = "Distinction"
    elif total >= 65:
        grade = "Merit"
    elif total >= 50:
        grade = "Pass"
    else:
        grade = "Fail"

    # write score and grade to file
    outString = str(total) + " " + grade + "\n"
    theFile.write(outString)

theFile.close()

print ("Grade file has been created")
```

19 of 23

Type MS · Read and write text files line by line

Answer 6 of 6 · Question 5 part b · 8 marks

2025 JUN · Paper 02 · extract 4 of 4 · find part b in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark